

Supply Chain Dashboard

Power BI Portfolio Case Study
Prepared by Zayna Thompson

Supply Chain Dashboard

Executive Overview

3K

Total Orders

\$274.70

Average Price Per Unit

14.96

Average Delivery Time(Days)

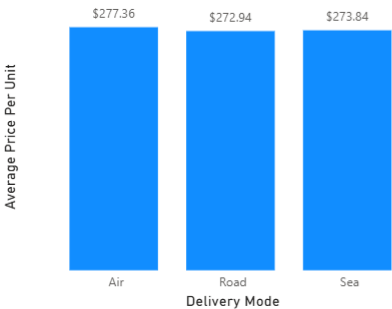
4.99%

Average Defect Rate

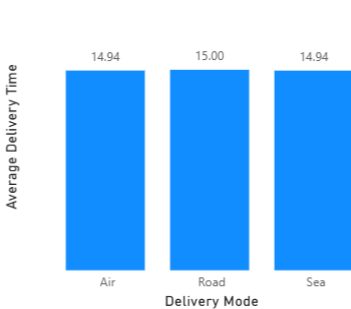
0.51

Average Supplier Reliability

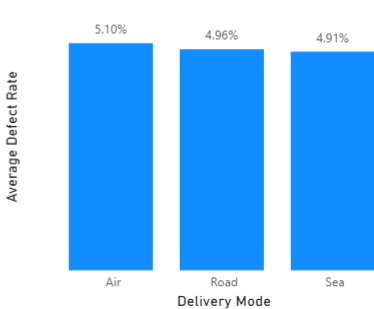
Average Price Per Unit by Delivery Mode

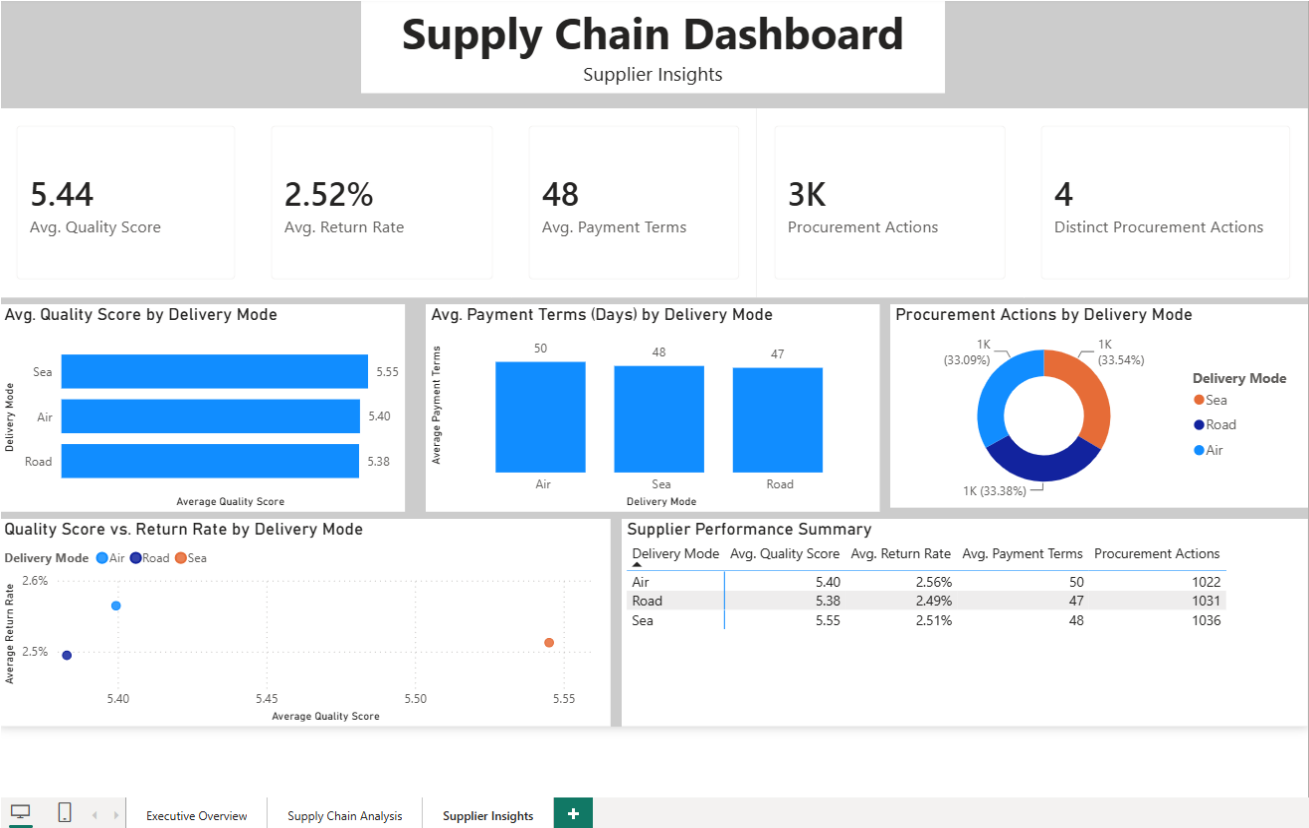
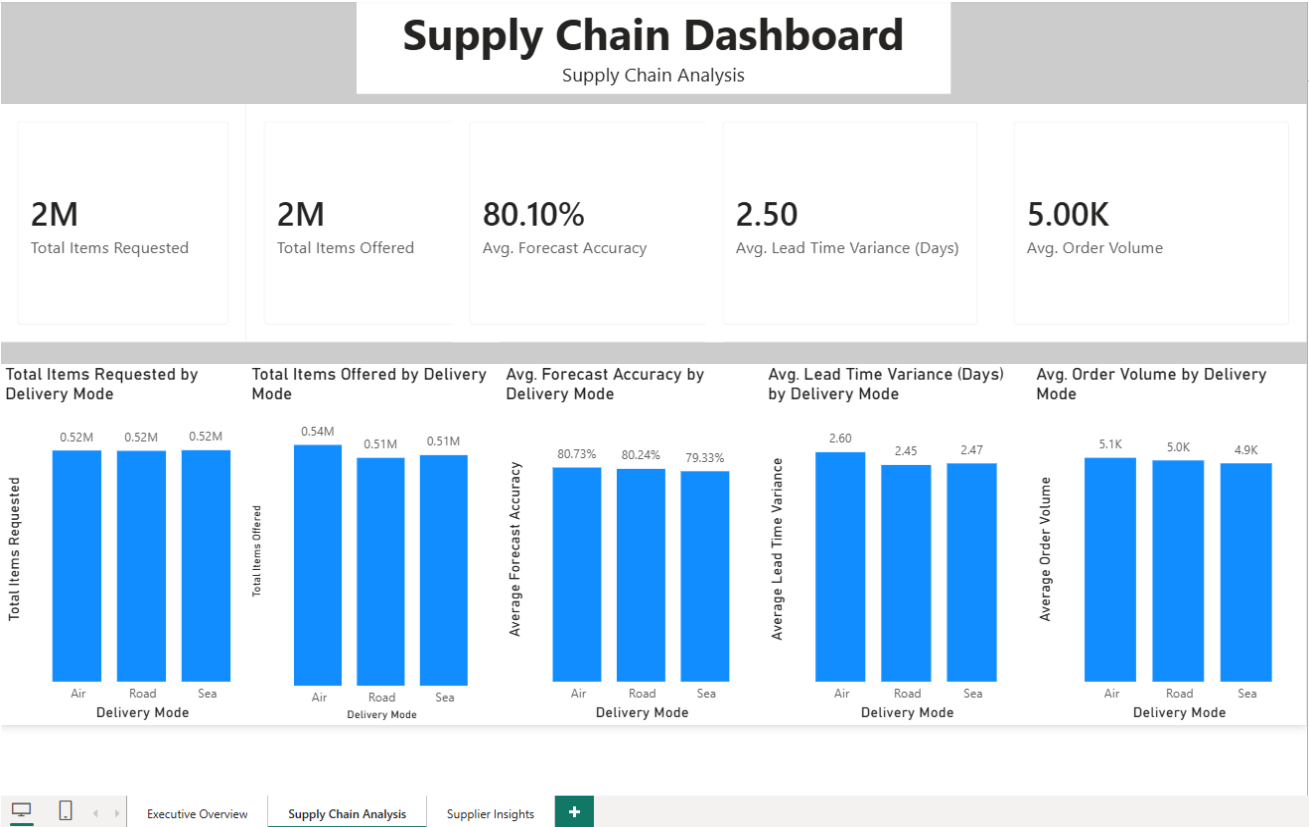


Average Delivery Time (Days)



Average Defect Rate by Delivery Mode





Executive Summary

This interactive Supply Chain Dashboard was developed using PostgreSQL, SQL, Power Query, DAX, and Power BI to transform operational supply chain data into executive-ready business intelligence. The solution delivers a centralized view of procurement, supplier performance, inventory movement, delivery operations, and quality metrics to support strategic decision-making.

Business Problem

Supply chain leaders often manage procurement, supplier performance, and logistics across disconnected reports, making it difficult to monitor operational efficiency, identify risks, and make timely business decisions.

Project Objectives

- Centralize supply chain KPIs.
- Evaluate supplier performance and procurement activity.
- Analyze delivery performance, forecasting accuracy, quality, and defect rates.
- Support executive decision-making through interactive reporting.

Challenge

Imported a 3,000+ row supply chain dataset into PostgreSQL to prepare data for Power BI analysis. During the import process, a schema mismatch prevented the CSV from loading successfully.

Approach

- Designed the PostgreSQL table schema to match the dataset.
- Debugged the import error by comparing the CSV structure to the database table.
- Corrected the schema and successfully imported the data.
- Queried the data using SQL to prepare it for analysis.
- Connected PostgreSQL directly to Power BI for visualization.

Tools & Technologies

PostgreSQL • SQL • Power BI • Power Query • DAX

Data Preparation

Data was imported into PostgreSQL, validated, queried using SQL, connected directly to Power BI, transformed in Power Query, modeled using relationships, and enhanced with DAX measures to produce interactive executive dashboards.

Dashboard Highlights

Executive KPI overview, delivery performance analysis, supplier insights, procurement metrics, forecast accuracy, lead-time variance, quality and return-rate analysis, interactive filtering, and executive reporting.

Key Insights

- Delivery performance remained consistent across transportation modes.
- Supplier quality scores were stable while return rates remained low.

- Forecast accuracy averaged approximately 80%, providing a strong planning baseline.
- Procurement activity was evenly distributed across delivery methods.

Business Recommendations

- Monitor forecast accuracy to improve inventory planning.
- Continue tracking supplier quality and defect rates.
- Review lead-time variance to reduce delivery risk.
- Use executive dashboards during operational review meetings.

Skills Demonstrated

PostgreSQL • SQL • Data Import & Validation • Data Cleaning • Troubleshooting • Power BI • Power Query • DAX • Data Modeling • Business Intelligence • Dashboard Development

Reflection

This project strengthened my ability to build an end-to-end analytics solution by troubleshooting database imports, designing PostgreSQL schemas, querying operational data with SQL, transforming data in Power Query, creating DAX measures, and delivering executive-ready Power BI dashboards.